

CONTRA
CONSULTATION AND TRAINING

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اونيفرسيتي مليسيا قهڠ
UNIVERSITI MALAYSIA PAHANG
Fakulti Teknologi Kejuruteraan
Pembuatan dan Mekatronik

Machine Learning and A.I Training Modules

Beginner

Supervised and Unsupervised Machine Learning

(Slot 3 & 4)

24th May 2023

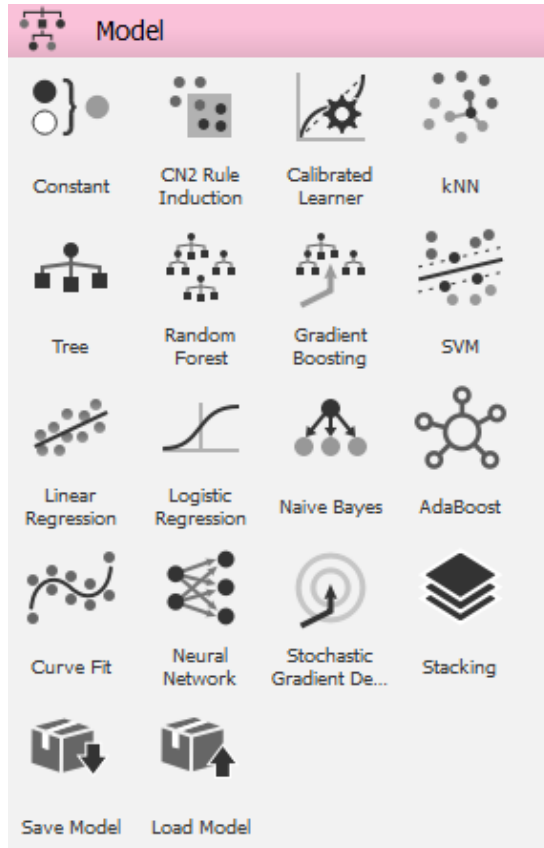


Supervised Machine Learning

- Supervised learning, also known as supervised machine learning, is defined by its use of labeled datasets to train algorithms to classify data or predict outcomes accurately.
- As input data is fed into the model, the model adjusts its weights until it has been fitted appropriately. This occurs as part of the cross validation process to ensure that the model avoids overfitting or underfitting.
- Supervised learning helps organizations solve a variety of real-world problems at scale, such as classifying spam in a separate folder from your inbox.
- Some methods used in supervised learning include neural networks, naïve bayes, linear regression, logistic regression, random forest, and support vector machine (SVM).

Supervised Machine Learning

- Keywords:
 - Labeled datasets
 - To train algorithms
 - To classify data or predict outcomes accurately



Supervised Machine Learning



File

File - Orange

Source

☒ File: iris.tab ☐ URL:

File Type

Automatically detect type

Info

Iris flower dataset
Classical dataset with 150 instances of Iris setosa, Iris virginica and Iris versicolor.

150 instances
4 features (no missing values)
Classification; categorical class with 3 values (no missing values)
0 meta attributes

Columns (Double click to edit)

	Name	Type	Role	Values
1	sepal length	N numeric	feature	
2	sepal width	N numeric	feature	
3	petal length	N numeric	feature	
4	petal width	N numeric	feature	
5	iris	C categorical	target	Iris-setosa, Iris-versicolor, Iris-virginica

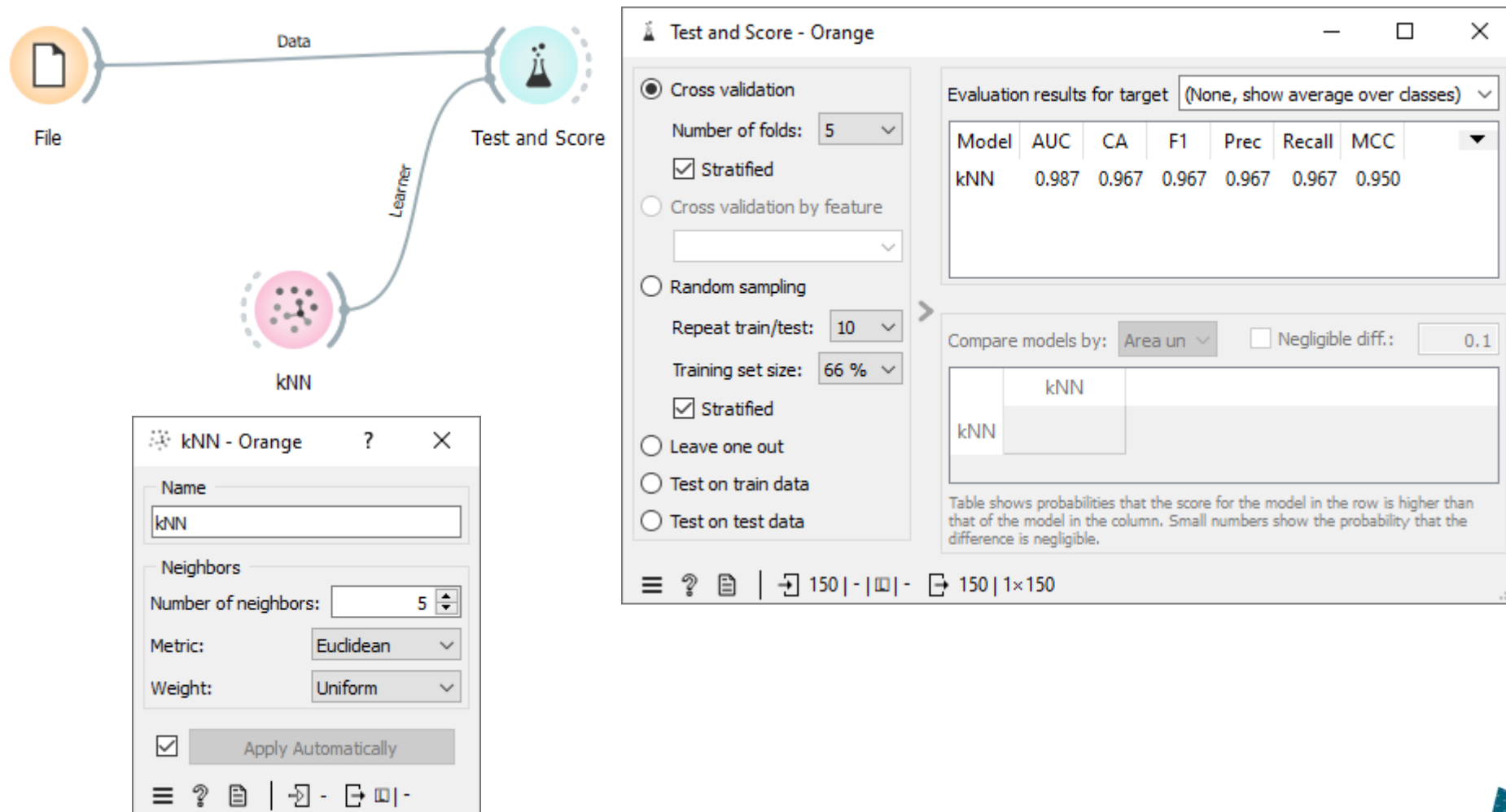
Reset Apply

Browse documentation datasets

150

- Use iris.tab data

Supervised Machine Learning: kNN



Supervised Machine Learning: SVM

SVM - Orange

Name: SVM

SVM Type

☒ SVM Cost (C): 1.00

Regression loss epsilon (ϵ): 0.10

☐ v-SVM Regression cost (C): 1.00

Complexity bound (v): 0.50

Kernel

☐ Linear Kernel: $\exp(-g|x-y|^2)$

☐ Polynomial g: auto

☒ RBF

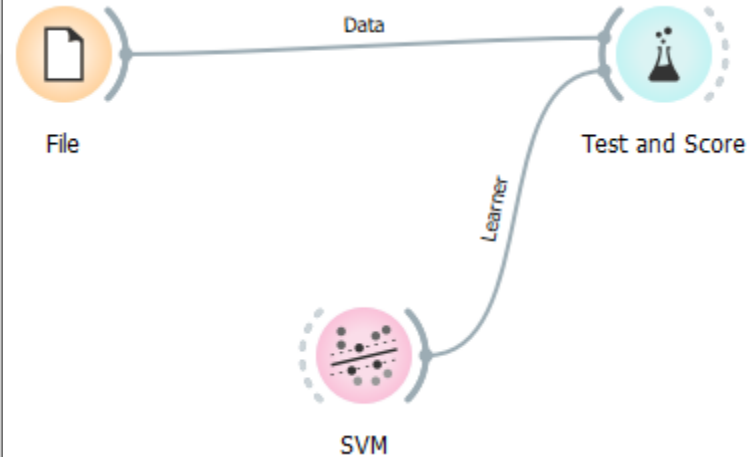
☐ Sigmoid

Optimization Parameters

Numerical tolerance: 0.0010

☒ Iteration limit: 100

☒ Apply Automatically



Test and Score - Orange

☒ Cross validation

Number of folds: 5

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 66 %

☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Evaluation results for target: (None, show average over classes)

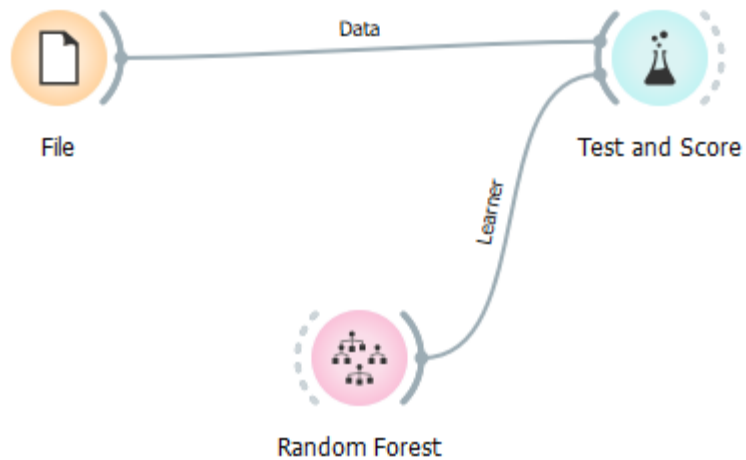
Model	AUC	CA	F1	Prec	Recall	MCC
SVM	0.997	0.960	0.960	0.960	0.960	0.940

Compare models by: Area un ☐ Negligible diff.: 0.1

	SVM
SVM	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Supervised Machine Learning: RF



Random Forest - Orange

Name: Random Forest

Basic Properties

Number of trees: 10

☐ Number of attributes considered at each split: 5

☐ Replicable training

☐ Balance class distribution

Growth Control

☐ Limit depth of individual trees: 3

☒ Do not split subsets smaller than: 5

☒ Apply Automatically

Test and Score - Orange

☒ Cross validation

Number of folds: 5

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 66 %

☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Evaluation results for target: (None, show average over classes)

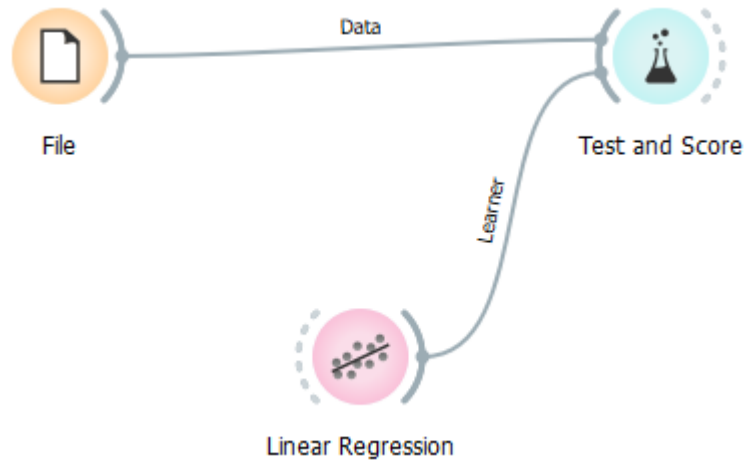
Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.995	0.960	0.960	0.960	0.960	0.940

Compare models by: Area un ☐ Negligible diff.: 0.1

Model	Random ...
Random Forest	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Supervised Machine Learning: LR



Linear Regression - Orange

Name: Linear Regression

Parameters: ☒ Fit intercept (unchecking it fixes it to zero)

Regularization: ☒ No regularization Regularization strength:
 ☐ Ridge regression (L2) Alpha: 0.0001
 ☐ Lasso regression (L1) Elastic net mixing: L1 L2
 ☐ Elastic net regression 0.50 : 0.50

☒ Apply Automatically

Test and Score - Orange

☒ Cross validation Number of folds: 5
 ☐ Stratified
 ☐ Cross validation by feature
 ☐ Random sampling Repeat train/test: 10
 Training set size: 66 %
 ☒ Stratified
 ☐ Leave one out
 ☐ Test on train data
 ☐ Test on test data

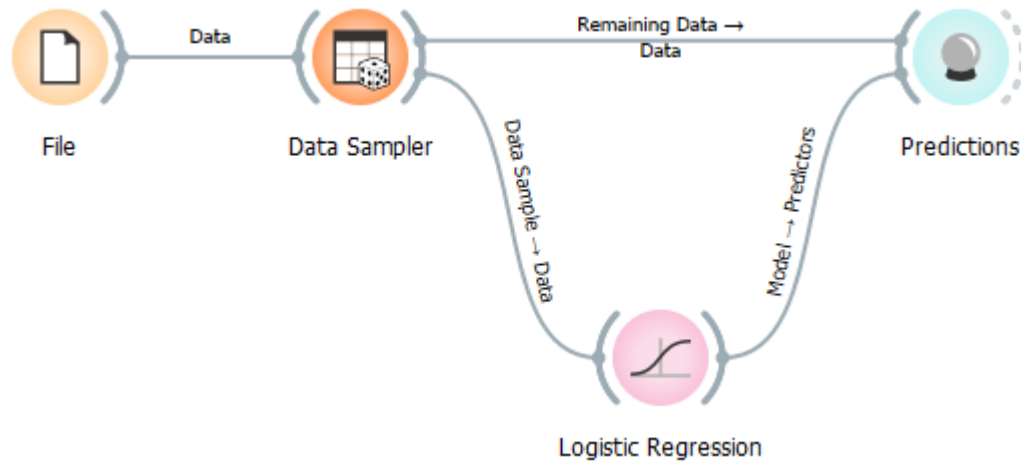
Model	MSE	RMSE	MAE	R2
Linear Regression	23.490	4.847	3.388	0.722

Compare models by: Mean sc Negligible diff.: 0.1

Model	Linear Re...
Linear Regression	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Supervised Machine Learning: Logistic Regression



Logistic Regression

Name: Logistic Regression

Regularization type: Ridge (L2)

Strength: Weak Strong
C=1

☐ Balance class distribution

☒ Apply Automatically

Predictions - Orange

Show probabilities for: Iris-setosa ☒ Show classification errors [Restore Original Order](#)

	Logistic Regression	error	iris	sepal length	sepal width	petal length	petal width
1	0.32 → Iris-versic...	0.322	Iris-versicolor	5.1	2.5	3.0	1.1
2	0.00 → Iris-virginica	0.041	Iris-virginica	6.9	3.2	5.7	2.3
3	0.00 → Iris-virginica	0.575	Iris-versicolor	6.0	2.7	5.1	1.6
4	0.00 → Iris-virginica	0.278	Iris-virginica	6.1	2.6	5.6	1.4
5	0.00 → Iris-virginica	0.017	Iris-virginica	7.7	3.0	6.1	2.3
6	0.04 → Iris-versic...	0.091	Iris-versicolor	5.5	2.5	4.0	1.3
7	0.98 → Iris-setosa	0.024	Iris-setosa	4.4	2.9	1.4	0.2
8	0.99 → Iris-setosa	0.010	Iris-setosa	4.3	3.0	1.1	0.1
9	0.00 → Iris-versic...	0.511	Iris-virginica	6.0	2.2	5.0	1.5
10	0.00 → Iris-virginica	0.069	Iris-virginica	7.2	3.2	6.0	1.8

☒ Show performance scores Target class: (Average over classes)

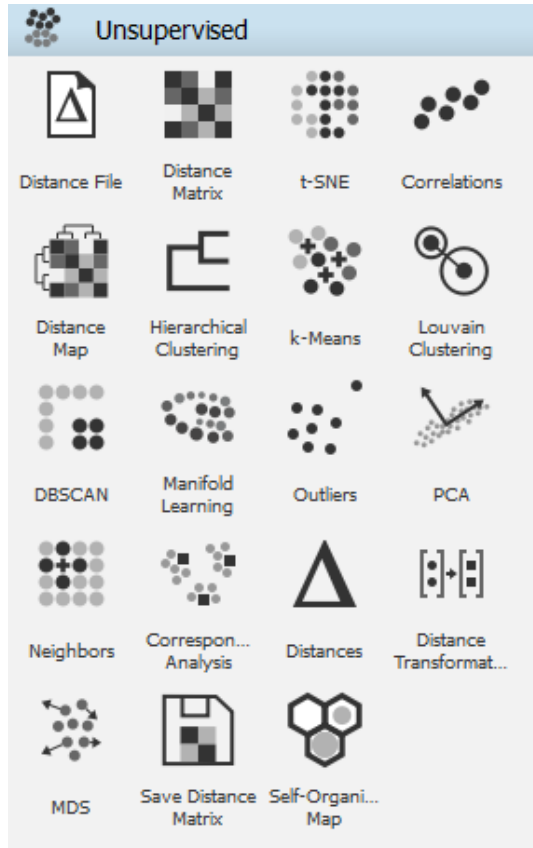
Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.996	0.950	0.950	0.951	0.950	0.924

TAKE A BREAK, HAVE A SEAT

Unsupervised Machine Learning

- Unsupervised learning, also known as unsupervised machine learning, uses machine learning algorithms to analyze and cluster unlabeled datasets.
- These algorithms discover hidden patterns or data groupings without the need for human intervention. This method's ability to discover similarities and differences in information make it ideal for exploratory data analysis, cross-selling strategies, customer segmentation, and image and pattern recognition.
- It's also used to reduce the number of features in a model through the process of dimensionality reduction. Principal component analysis (PCA) and singular value decomposition (SVD) are two common approaches for this.
- Other algorithms used in unsupervised learning include neural networks, k-means clustering, and probabilistic clustering methods.

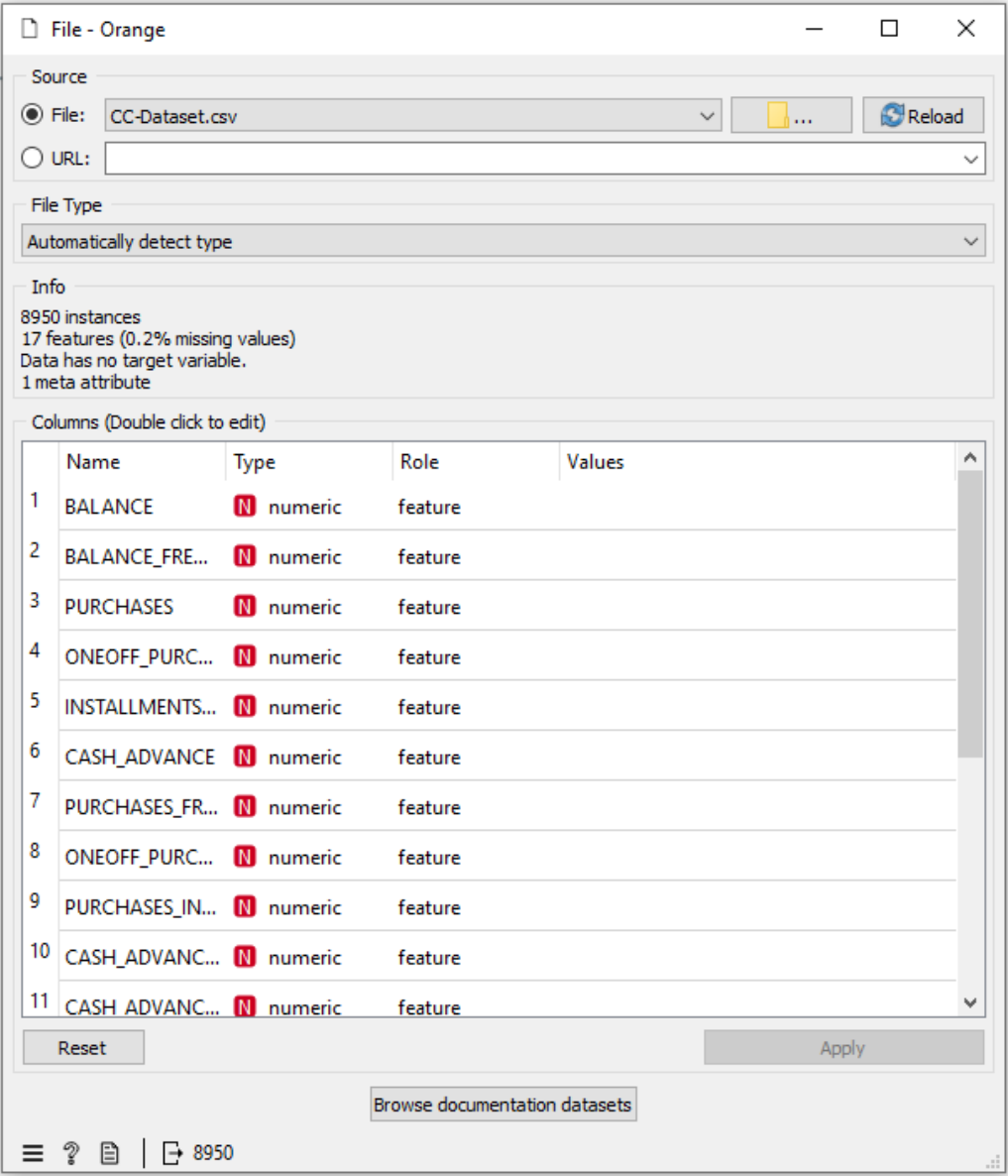
Unsupervised Machine Learning



- Keywords:
 - Unlabeled datasets.
 - To analyze and cluster unlabeled datasets.
- These algorithms discover hidden patterns or data groupings without the need for human intervention.
- This method's ability to discover similarities and differences in information make it ideal for exploratory data analysis, cross-selling strategies, customer segmentation, and image and pattern recognition.
- It's also used to reduce the number of features in a model through the process of dimensionality reduction. Principal component analysis (PCA) and singular value decomposition (SVD) are two common approaches for this.
- Other algorithms used in unsupervised learning include neural networks, k-means clustering, and probabilistic clustering methods.

Unsupervised Machine Learning

File

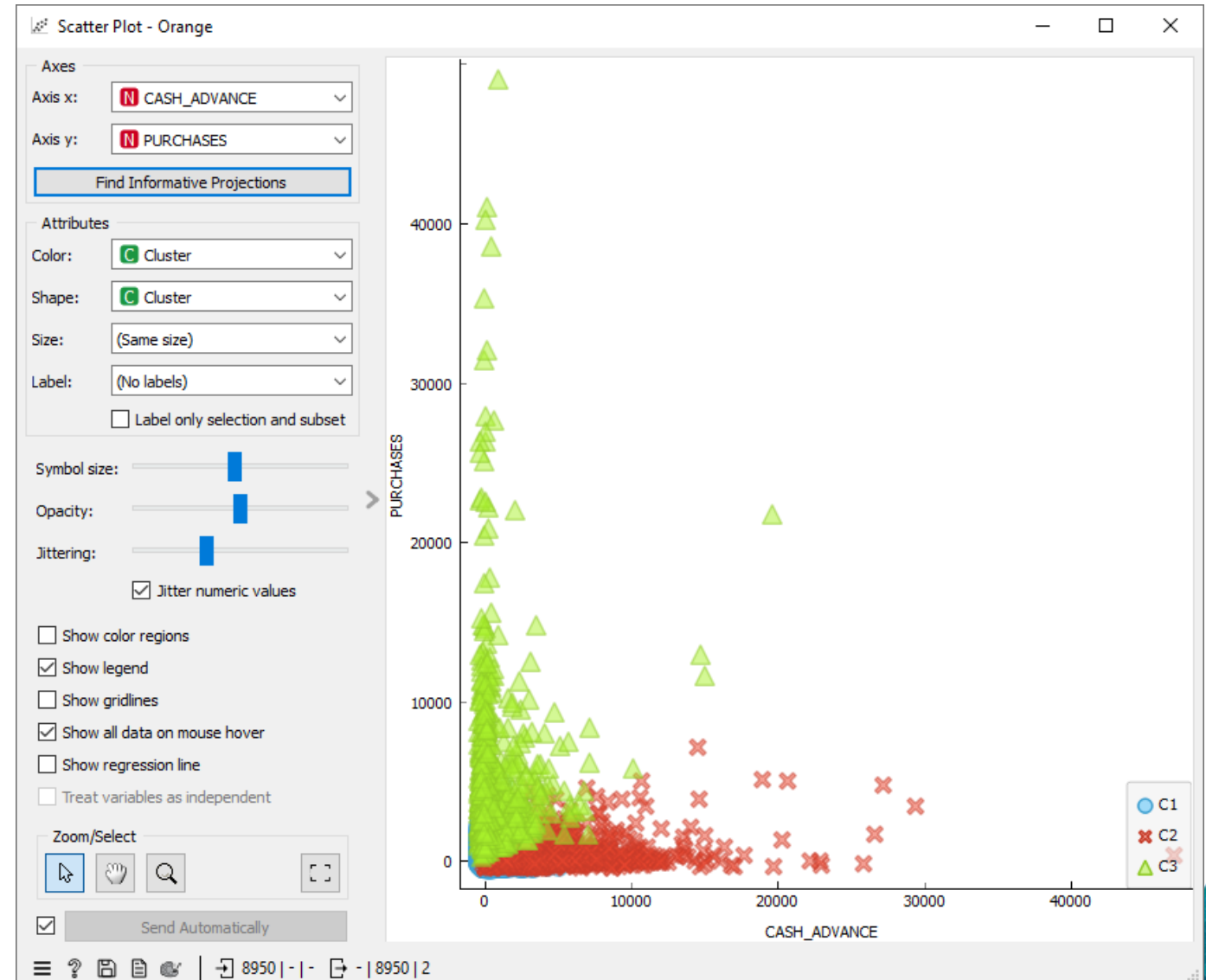
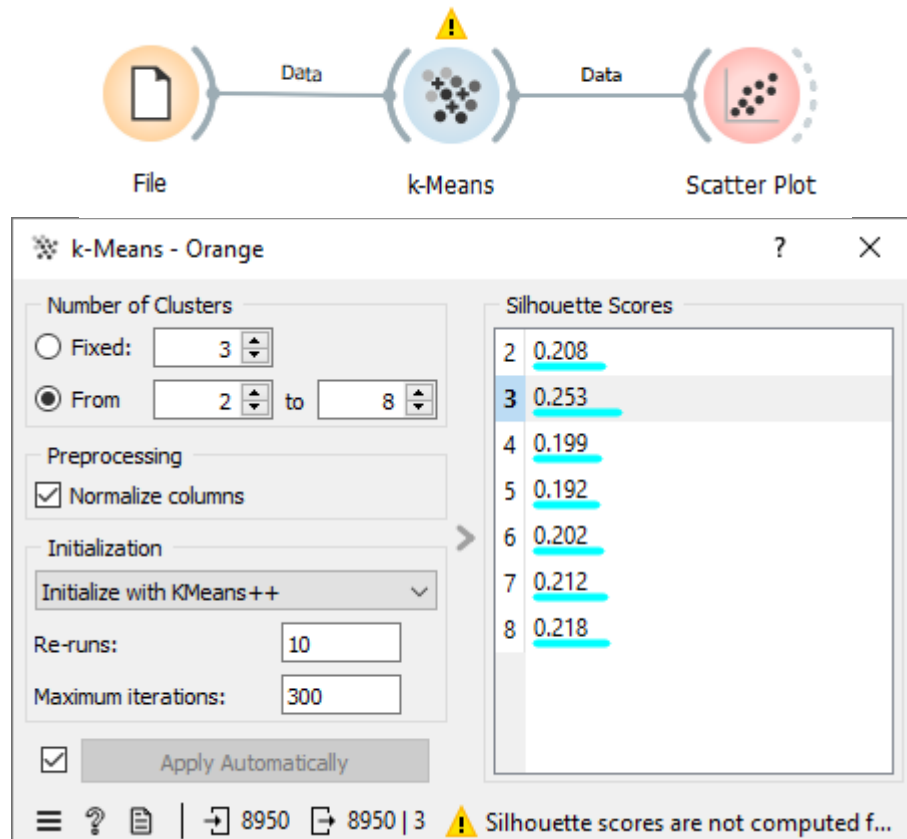


The screenshot shows the 'File - Orange' window. The 'Source' section has 'File: CC-Dataset.csv' selected. The 'File Type' is set to 'Automatically detect type'. The 'Info' section displays: '8950 instances', '17 features (0.2% missing values)', 'Data has no target variable.', and '1 meta attribute'. The 'Columns (Double click to edit)' table lists 11 features, all of which are numeric and have a role of 'feature'. At the bottom, there are 'Reset' and 'Apply' buttons, and a status bar showing '8950' instances.

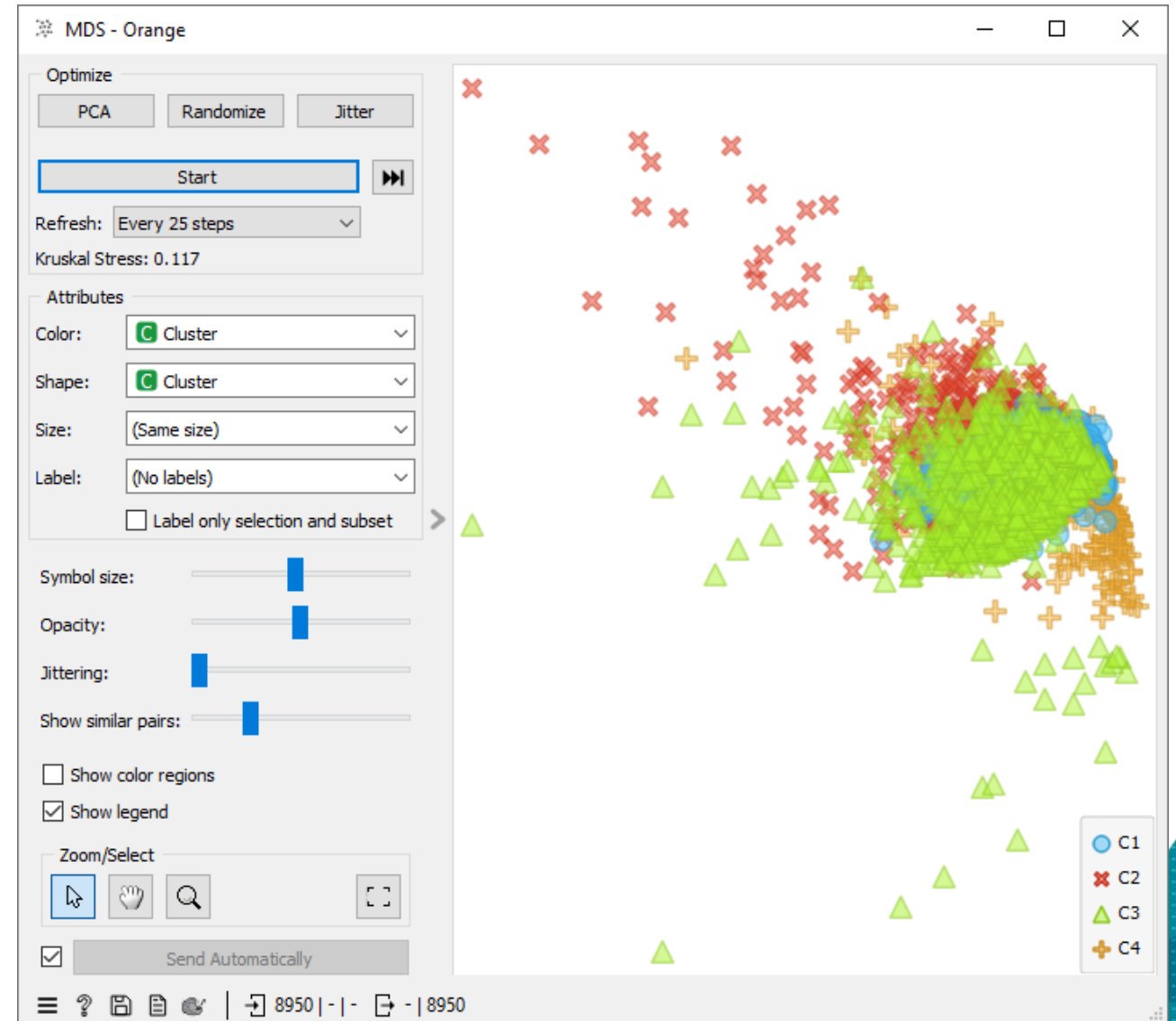
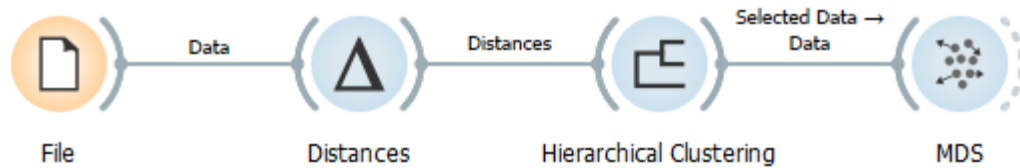
	Name	Type	Role	Values
1	BALANCE	N numeric	feature	
2	BALANCE_FRE...	N numeric	feature	
3	PURCHASES	N numeric	feature	
4	ONEOFF_PURC...	N numeric	feature	
5	INSTALLMENTS...	N numeric	feature	
6	CASH_ADVANCE	N numeric	feature	
7	PURCHASES_FR...	N numeric	feature	
8	ONEOFF_PURC...	N numeric	feature	
9	PURCHASES_IN...	N numeric	feature	
10	CASH_ADVANC...	N numeric	feature	
11	CASH ADVANC...	N numeric	feature	

- Use CC-Dataset.csv data

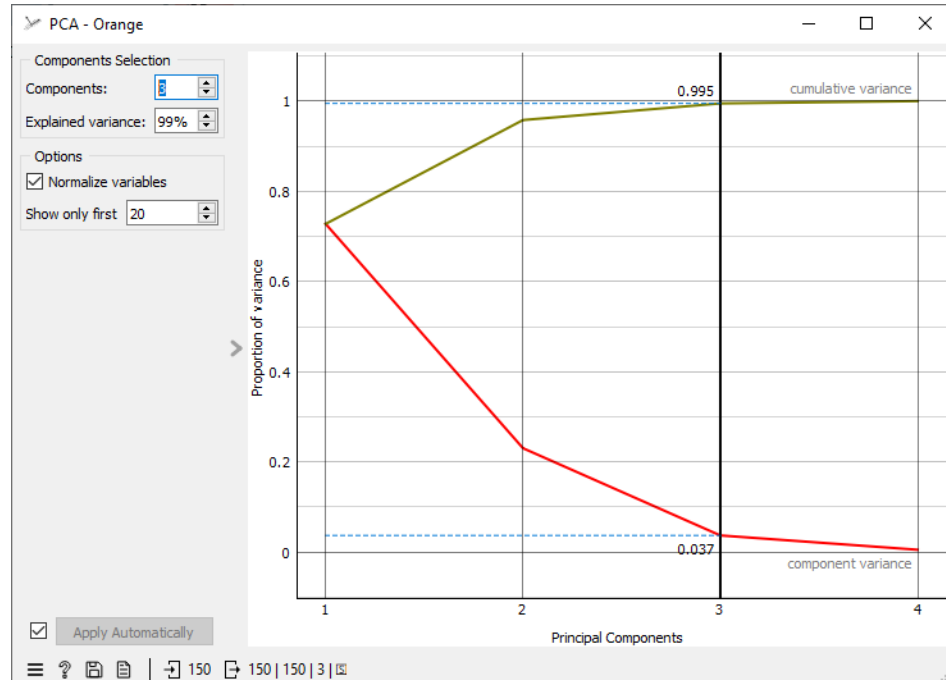
Unsupervised Machine Learning: k-Means



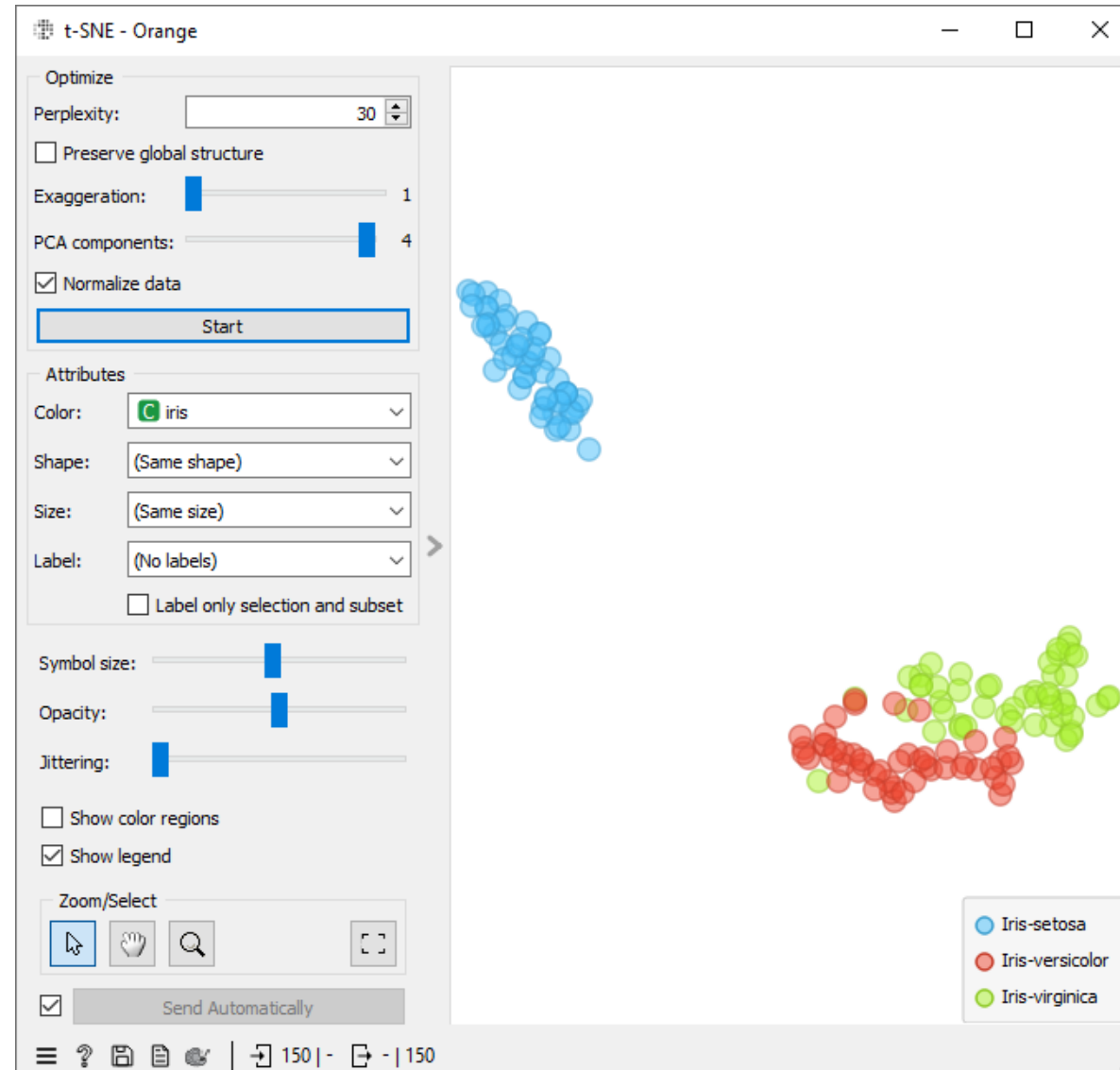
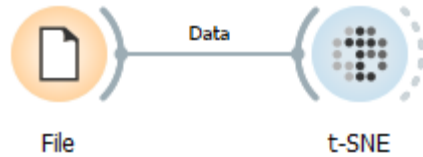
Unsupervised Machine Learning: Hierarchical Clustering (HCA)



Unsupervised Machine Learning: PCA



Unsupervised Machine Learning: t-SNE



THANK YOU